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Palæomagnetic results and palæointensity of Late Cretaceous Madagascan basalt

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ABSTRACT—For a palæomagnetic study of the Late Cretaceous flood basalts of Madagascar (age ~88 Ma), 144 oriented drill cores from 15 sites located in the northwestern part of the island were sampled. Despite the very common occurrence of secondary isothermal remanent magnetisation related to lightning, the direction of the primary characteristic remanent magnetisation could, in many cases, be isolated using alternating field demagnetisation. Combining the palæodirectional results from this study with previously published results, the authors obtain a well-defined palæomagnetic pole for Madagascar with coordinates 67.8°S, 48.5°E ($A_{95} = 3.7°$, $K = 34$, $N = 44$). This pole is in good agreement with the synthetic apparent polar wander path (APWP) for Africa. Furthermore, Bingham statistics provided a long axis lying along the APWP indicating that the duration of volcanic activity was sufficiently long to record apparent polar wander. Palæosecular variation, $S = 13.9\ (^{+2.4}/_{-1.8})$, is slightly smaller than expected for this time interval and latitudinal band.

The absolute palæointensity, using the Thellier technique, could be obtained only from a single cooling unit, yielding a palæofield of $61.5 \pm 5.5\ \mu T$. The corresponding virtual dipole moment of 10.7×10^{22} A m^2 is somewhat higher than previously published results from the Cretaceous normal superchron; but, obviously, secular variation is not averaged. © 2001 Elsevier Science Limited. All rights reserved.

RÉSUMÉ—Une étude paléomagnétique a été effectuée sur les roches basaltiques Crétacé supérieur de Madagascar (~88 Ma); 144 carottes orientées (15 sites) provenant de la partie nord-ouest de Madagascar ont été prélevées. Malgré l'occurrence fréquente d'une aimantation secondaire due à la foudre, la direction de l'aimantation rémanente caractéristique a pu être isolée, dans de nombreux cas, après désaimantation par champ alternatif. Nos résultats paléodirectionnels ont été combinés avec les données fiables précédemment publiées, et un pôle paléomagnétique de bonne qualité, dont les coordonnées sont 67.8°S, 48.5°E ($A_{95} = 3.7°$, $K = 34$, $N = 44$), a été obtenu. Ce pôle est en bon accord avec le courbe synthétique de dérive apparente des pôles de l'Afrique. De plus, le grand axe de l'ellipse obtenue par la statistique de Bingham est dirigé selon la courbe de dérive, ce qui semble indiquer que l'activité volcanique a été suffisamment longue pour avoir enregistré la dérive apparente du pôle. Le taux de variation paléoséculaire, $S = 13.9\ (^{+2.4}/_{-1.8})$, est légèrement inférieur à la valeur attendue pour cet intervalle de temps et pour cette latitude.

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Une seule coulée basaltique a permis une détermination de paléointensité par la méthode de Thellier, avec une valeur du champ ancien de 61.5 ± 5.5 μT. Le moment du dipôle virtuel correspondant, 10.7×10^{22} A m^2, est supérieur à ceux précédemment publiés pour la longue période normale du Crétacé, mais il est évident que la variation séculaire n'a pas été moyennée. © 2001 Elsevier Science Limited. All rights reserved.

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INTRODUCTION

While there have been no previous attempts to obtain palæointensity determinations from the Late Cretaceous basalts of Madagascar, palæodirectional results have been published by Roche and Cattala (1959), Nairn (1964), Andriamirado and Roche (1969), Andriamirado (1971, 1976), Storetvedt *et al.* (1992) and Torsvik *et al.* (1998). The most extensive sampling and study is the one accomplished by Andriamirado (1971, 1976), and his palæodirectional data have been used in the most recent updates of both synthetic APWPs for Africa, Eurasia, North America and India (Besse and Courtillot, 1991; Besse *et al.*, 1996), as well as the estimation of palæosecular variation (McFadden *et al.*, 1991). Interpretations of the results of Andriamirado (1971, 1976) are, however, spoiled by the inclusion in his study of palæomagnetic sites with poor statistics and imprecise K/Ar ages, leading to an erroneous grouping of data into different age groups. New geochronological results (Storey *et al.*, 1995; Torsvik *et al.*, 1998) indicate that the Madagascan Cretaceous volcanism had a relatively short duration. Here, therefore, the authors have chosen to combine the new data from their study with previously published results from the Cretaceous volcanics of Madagascar in order to create a single palæomagnetic directional data set in a manner similar to what has been done for the Deccan Traps (Vandamme *et al.*, 1991). They, hereby, produce a rather large palæomagnetic data set, from which is generated a more reliable palæomagnetic pole and a better estimate of the palæosecular variation, than could have been obtained from the separate studies.

The long-term geological evolution of the geodynamo has been the focus of great interest in recent years. Much of the attention has been directed towards the Cretaceous normal superchron (CNS) with a duration of about 35 Ma, from 118 to 83 Ma according to Cande and Kent (1995) (see also Prévot *et al.*, 1990; Larson and Olson, 1991; Pick and Tauxe, 1993; McFadden and Merrill, 1995). This interest was fueled by the suggested link between the strength of the geomagnetic field and the processes controlling reversal frequency. However, the few already published palæointensity determinations from the CNS do not point to anomalously high (or low) palæointensities, but rather suggest that the processes controlling palæointensity and reversal frequency are decoupled (McFadden and Merrill, 1986; Prévot *et al.*, 1990; Pick and Tauxe, 1993). With only 12 reliable palæointensity determinations for the entire CNS (Perrin and Shcherbakov, 1997), much more data are needed in order to draw any firm conclusions. The original goal for the sampling of the Madagascan basalts in this study (extruded around 88 Ma) was to obtain additional palæointensity determinations within the CNS. Unfortunately, the Madagascan basalts are not very suited for that type of experiment and only one flow allowed palæointensity determinations.

GEOLOGY AND SAMPLING

Palæomagnetic results from sedimentary rocks of the Karoo Supergroup of Madagascar (Embleton and McElhinny, 1975; McElhinny *et al.*, 1976; Razafindrazaka *et al.*, 1976) suggest that, prior to the Gondwana breakup, Madagascar was in a northern position adjacent to Kenya, Tanzania and southern Somalia. Interpretation of the Mesozoic series of sea floor spreading magnetic anomalies in the western Somali Basin indicates that the southward directed drift of Madagascar relative to Africa began in the Jurassic quiet zone (older than 162 Ma) and ceased around Chron M9 (~129 Ma) (Rabinowitz *et al.*, 1983) or possibly Chron M0 (~118 Ma) (Ségoufin and Patriat, 1980).

In Late Cretaceous times, greater India broke away from Madagascar with voluminous igneous activity accompanying the continental break-up. A geochemical study of the Madagascan flood basalts suggests that the Marion hot spot was an important source of melts at the southern end of the island (Mahoney *et al.*, 1991; Storey *et al.*, 1997). This is also supported by a palæogeographic reconstruction placing Madagascar in the near vicinity of the Marion hot spot around Late Cretaceous times (Storey *et al.*, 1995; Torsvik *et al.*, 1998). Today, the Late Cretaceous volcanic and intrusive rocks of Madagascar crop out semi-continuously along the east coast (Côte Centre Nord-Est, Côte Sud-Est), in the west (Majunga, Mailaka, Mangoky-Onilahy) and in the south (Androy) of the island (Fig. 1). The rocks include basalt and rhyclite flows, dolerite dykes and microgabbroic intrusions (Boast and Nairn, 1982). Seventeen $^{40}Ar/^{39}Ar$ datings of

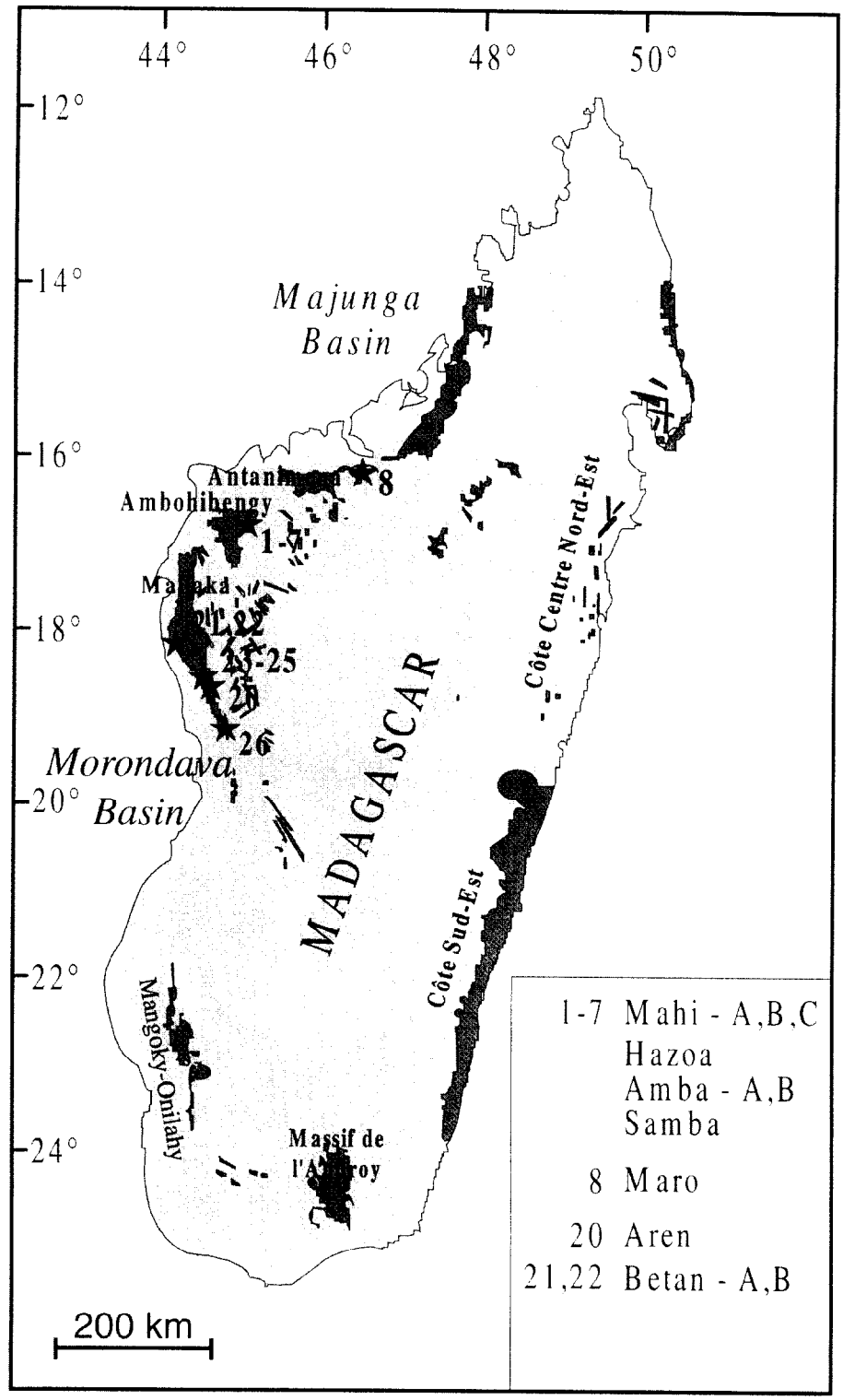


Figure 1. *Simplified geological map of Madagascar with the Late Cretaceous flood basalts shown as dark areas. The site numbers are the same as in Table 1. ★: The locations of the sites sampled in this study.*

the volcanics give a mean age of 87.6 ± 0.6 Ma (Storey *et al.*, 1995), with ages falling between 90.1 and 83.7 Ma. Though the position of the dated rocks in the stratigraphy is unknown, the fact that they spread out over all the volcanic provinces tends to indicate that the duration of volcanism on the island would be no more than 6 Ma (Storey *et al.*, 1995). Hence, the entire volcanic activity took place during the CNS.

In this study, the sampling of the Late Cretaceous basalts was confined to outcrops at the Antanimena, Mailaka and Ambohibengy localities in the north-western part of the island. The geographic location of the sites is shown in Fig. 1, and the coordinates are given in Table 1. All cores were drilled directly in the field with a portable gasoline driven drill and orientated using both magnetic and sun compasses. No evidence for local tectonic disturbance could be seen in the field.

ROCK MAGNETIC INVESTIGATIONS

Prior to the palæomagnetic work, some rock magnetic experiments were carried out to evaluate the magnetic mineralogy and suitability of the samples for palæointensity experiments. First of all, the magnetic viscosity index was determined for all samples (Thellier and Thellier, 1944). Most of the samples were stable with very low viscosity indexes, <3%.

Susceptibility versus temperature

Temperature dependence of initial susceptibility $\kappa(T)$ was measured on one or two samples from each site. Measurements were done under vacuum (better than 10^{-3} mbar) using a Bartington susceptibility meter equipped with a furnace. Exhibited behaviours can be described as follows:

i) close to reversible $\kappa(T)$ curves with single Curie point around 150–250°C (Fig. 2a) (the difference in heating and cooling curves can be partly attributed to a difference between heating and cooling rates);

ii) curves with the heating curve having two or three peaks in the 150–450°C interval and which are not reversible in cooling (Fig. 2b);

iii) $k(T)$ curves which are reversible up to 400°C (see first cycle in Fig. 2c) while the full $\kappa(T)$ cycle up to 600°C is irreversible (Fig. 2c); and

iv) close to reversible $\kappa(T)$ curves (Fig. 2d) with a Curie temperature of magnetite, around 560–580°C.

It can be noted that none of the $\kappa(T)$ curves indicate the presence of a major amount of secondary magnetic minerals like maghemite, lepidocrocite or goethite typical for low-temperature oxidation of Fe oxides in weathered regions. The simple $\kappa(T)$ curves of Fig. 2a, d might be identified as primary titanomagnetite (TM60) and pure magnetite (TM0), respectively. The more complex $\kappa(T)$ curves of Fig. 2b, c could be titanomagnetite with a Ti content lying in between TM60 and TM0, but some degree of maghemitisation cannot be excluded. An important remark is that only samples from Ambohibengy lava flows have the magnetite (TM0) $\kappa(T)$ curves (Fig. 2d). Samples from Antanimena and Mailaka localities, on the other hand, yield a diversity of $\kappa(T)$ curves (Fig. 2a–c).

Hysteresis

Magnetic hysteresis parameters were determined for 3–4 samples from each site using an alternating gradient force magnetometer. The saturation remanent magnetisation (J_{rs}), the saturation magnetisation (J_s) and the coercive force (B_c) were calculated after correction for the paramagnetic contribution. The coercivity of remanence (B_{cr}) was determined by applying progressively increasing backfields after saturation.

The ratios of hysteresis parameters are plotted on a Day plot (Day *et al.*, 1977) in Fig. 3. The Antanimena and Mailaka lava flows, which had varying $\kappa(T)$ curves (Fig. 2a–c), are closely grouped in the pseudo-single domain (PSD) region; while the Ambohibengy samples, which had close to pure magnetite $\kappa(T)$ curves (Fig. 2d), are seen to be scattered along the empirical power line for mixed single domain (SD) and multidomain (MD) magnetite ($J_{rs}/J_s = 0.5 \times (B_{cr}/B_r)^m$, $m = -1.13$) (Parry, 1982).

The reason for the difference in magnetic mineralogy between the Ambohibengy lava flows and the rest of the samples might be hydrothermal alteration of the samples during the Ambohibengy intrusion. As a result, the magnetic carriers of Ambohibengy lavas would be secondary pure magnetite with widely varying grain sizes. It was noted that very similar examples of hydrothermal alteration in lavas, creating secondary pure magnetite, have been previously reported, e.g. van der Voo *et al.* (1993) and Harlan *et al.* (1996). Samples of Antanimena and Mailaka lava flows yielded quite a diversity of mineralogical compositions, possibly due to more or less deuterically oxidised titanomagnetite, with grain sizes in the PSD range.

PALÆODIRECTIONAL RESULTS

To isolate the characteristic remanent magnetisation (ChRM), both step-wise alternating field (AF), and thermal and low temperature (LT) demagnetisation were performed. Thermal demagnetisation was done in vacuum (better than 10^{-4} mbar) using a temperature increment of 50°C. AF demagnetisation was performed using a laboratory built three-axis AF demagnetiser. LT demagnetisation was done by immersing the samples in liquid N_2 for one hour and then reheating samples to room temperature in zero field. All remanences were measured with a three-axis CTF cryogenic magnetometer, and the direction of the ChRM was calculated using standard principal component analysis (Kirschvink, 1980).

Scattered within site directions of the natural remanent magnetisation (NRM), high intensities of NRM (up to 1.1×10^{-2} A m^2 kg^{-1}) and very high Königsberger ratio (up to 120) indicate the presence

Table 1. Palæomagnetic data for five localities of Madagascan flood basalts

Locality	Site	Site Lat °N	Site Long °E	n/N	Inc°	Dec°	K	α_{95}°	VGP Lat °N	VGP Long °E
Ambohibengy	1. Mahi – A *	-16.88	45.10	0/7						
	2. Mahi – B *	-	-	0/4						
	3. Mahi – C *	-	-	0/5						
	4. Hazoa *	-16.90	45.10	0/6						
	5. Amba – A *	-16.87	45.12	6/7	-39.9	354.4	92	7.0	-82.2	86.1
	6. Amba – B *	-	-	6/7	-49.1	352.9	92	7.0	-75.4	70.0
	7. Samba *	-16.85	44.95	0/7						
Antanimena	8. Maro *	-16.23	46.57	10/10	-51.2	6.7	752	1.8	-73.3	26.5
	9. Ambatomainty **	-16.22	46.63	5/6	-53.0	8.0	167	6.0	-71.2	25.5
	10. Ampijoroe **	-16.23	46.50	4/5	-54.0	6.0	1435	2.0	-70.9	31.2
	11. Antanimavo **	-16.32	46.62	5/5	-50.0	346.0	334	4.0	-70.7	85.6
	12. Andrafinahoany **	-16.33	45.87	3/4	-62.0	360.0	711	5.0	-63.1	45.9
	13. Marovitsika **	-16.35	46.16	3/5	-54.0	3.0	385	6.0	-71.6	38.3
	14. Thopy-nord **	-16.45	45.83	3/4	-68.0	2.0	364	6.0	-55.4	43.6
	15. Mahavavy **	-16.52	46.67	3/3	-56.0	3.0	113	4.0	-69.8	39.7
	16. Andrafiambony **	-16.53	45.88	3/3	-74.0	4.0	2385	2.0	-46.3	43.0
	17. Besieky **	-16.58	45.83	6/9	-66.0	329.0	90	7.0	-49.4	77.6
	18. Ihopy-Sud **	-16.63	45.77	3/5	-60.0	7.0	516	5.0	-64.9	33.2
	19. Ambohitramboalambo **	-16.70	45.85	3/4	-57.0	347.0	597	5.0	-66.2	72.0
Mean	**12 sites (51 samples)**					***S*=9.4**	**42**	**6.7**	**-65.6**	**48.4**
						α_{1-3}=5.2	**α_{1-2}=6.7**	**θ=33.5**	**-65.8**	**48.1**
Mailaka	20. Aren *	-18.80	45.10	11/11	-58.9	349.1	311	2.6	-67.1	67.1
	21. Betan – A *	-18.20	44.18	12/13	-57.0	359.5	215	3.0	-70.6	45.5
	22. Betan – B *	-	-	0/5						
	23. Tsian – A *	-18.75	44.53	11/11	-51.4	5.1	400	2.3	-75.9	26.5
	24. Tsian – B *	-	-	9/9	-47.7	355.3	567	2.2	-79.1	66.8
	25. Tsian – C *	-	-	7/7	-50.7	356.9	1008	1.9	-77.0	56.4
	26. Beko *	-19.18	44.73	8/8	-57.0	10.0	693	2.1	-69.6	21.5
	27. Marifilaly **	-17.73	44.20	4/5	-58.0	355.0	235	6.0	-68.6	54.9
	28. Vohidroa **	-18.67	44.45	4/4	-66.0	330.0	667	3.0	-51.7	76.9
Mean	**8 sites (66 samples)**					***S*=8.4**	**55**	**7.5**	**-71.0**	**54.9**
						α_{1-3}=3.6	**α_{1-2}=7.8**	**θ=42.7**	**-71.1**	**54.6**
Côte Centre Nord-Est	29. Fénérive carrière **	-17.38	47.10	12/12	-65.0	9.0	79	5.0	-59.5	34.9
	30. Bemenaka **	-18.33	47.01	4/6	-62.0	7.0	285	5.0	-64.4	35.2
	31. Ambodivato-Sud **	-18.33	47.05	4/4	-60.0	8.0	967	4.0	-66.4	31.8
	32. Fanadrana **	-18.25	46.93	3/3	-67.0	15.0	356	6.0	-56.4	29.3
	33. Ambarimilambana **	-18.43	46.82	3/4	-64.0	23.0	2490	5.0	-56.7	16.9
	34. Brickaville **	-18.82	46.73	9/11	-61.0	13.0	167	4.0	-64.3	24.1
Mean	**6 sites (35 samples)**					***S*=3.8**	**213**	**4.6**	**-61.4**	**28.4**
						α_{1-3}=3.6	**α_{1-2}=4.5**	**θ=147.7**	**-61.4**	**28.4**
Côte Sud-Est	35. Andranomanitsy **	-20.62	48.27	5/5	-61.0	353.0	613	3.0	-67.8	62.1
	36. Anosimparihy **	-21.15	48.23	6/6	-62.0	350.0	546	3.0	-66.4	66.6
	37. Manatsinjo **	-21.25	48.32	4/4	-60.0	357.0	179	7.0	-70.2	55.0
	38. Manakara-Vohipeno A **	-22.16	47.97	4/8	-71.0	29.0	1485	2.0	-50.4	22.4
	39. Manakara-Vohipeno B **	-22.23	47.88	11/12	-55.0	22.0	135	4.0	-66.7	357.5
	40. Vohipeno **	-22.33	47.85	4/5	-69.0	13.0	217	6.0	-58.2	32.8
	41. Vohipeno-Farfangana **	-22.67	47.80	3/3	-54.0	13.0	1318	3.0	-73.6	6.9
	42. Farafangana **	-22.77	47.17	30/38	-62.0	356.0	86	3.0	-69.3	55.4
	43. Lopary **	-23.18	47.68	3/4	-60.0	353.0	381	6.0	-71.3	64.4
Mean	**9 sites (70 samples)**					***S*=8.6**	**43**	**7.9**	**-67.8**	**39.2**
						α_{1-3}=5.2	**α_{1-2}=7.9**	**θ=115.1**	**-67.9**	**39.5**
Mangoky-Onilahy	44. Tsiarimpioky **	-21.95	44.03	5/5	-61.0	340.0	1500	2.0	-63.8	79.2
	45. Miary-Sud **	-22.45	44.45	5/5	-56.0	330.0	181	6.0	-60.5	99.1
	46. Mamahapaha **	-22.50	44.22	5/5	-75.0	304.0	216	5.0	-36.5	72.9
	47. Andolofotsy **	-22.77	44.32	5/6	-59.0	321.0	138	6.0	-52.9	97.8
	48. Anapaly **	-22.87	44.20	6/6	-50.0	2.0	937	2.0	-81.9	31.9
	49. Manety **	-22.98	44.17	5/6	-51.0	7.0	1333	2.0	-79.3	10.2
	50. Vineta **	-23.05	44.27	11/12	-60.0	11.0	125	4.0	-69.9	9.5
	51. Anketa **	-23.12	44.28	6/6	-48.0	355.0	1000	2.0	-82.6	80.4
	52. Vatolatsaka **	-23.32	44.28	3/3	-58.0	351.0	1000	4.0	-72.9	68.8
Mean	**9 sites (51 samples)**					***S*=13.3**	**18**	**12.2**	**-69.1**	**72.6**
						α_{1-3}=6.2	**α_{1-2}=13.3**	**θ=23.6**	**-69.1**	**71.9**
Mean Madagascar	**44 sites**					***S*=10.1**	**34**	**3.7**	**-67.8**	**48.5**
						α_{1-3}=3.3	**α_{1-2}=4.1**	**θ=66.4**	**-67.9**	**48.1**

*: This study; **: Andriamirado (1976); n/N: number of used/treated samples; Inc.: inclination of the mean site ChRM; Dec.: declination of the mean site ChRM; κ: precision parameter of the Fisher statistics; α_{95}: confidence cone of the Fisher statistics; VGP Lat: latitude of the Virtual Geomagnetic Pole; VGP Long: longitude of the Virtual Geomagnetic Pole; *S*: angular standard deviation; α_{1-2}: maximum (long) principal axis of the confidence ellipse of the Bingham statistics; α_{1-3}: minimum principal axis; θ: azimuth of the long axis (angle between long axis and a direction to the North Pole). Mean Fisher and Bingham parameters are listed for each locality site.

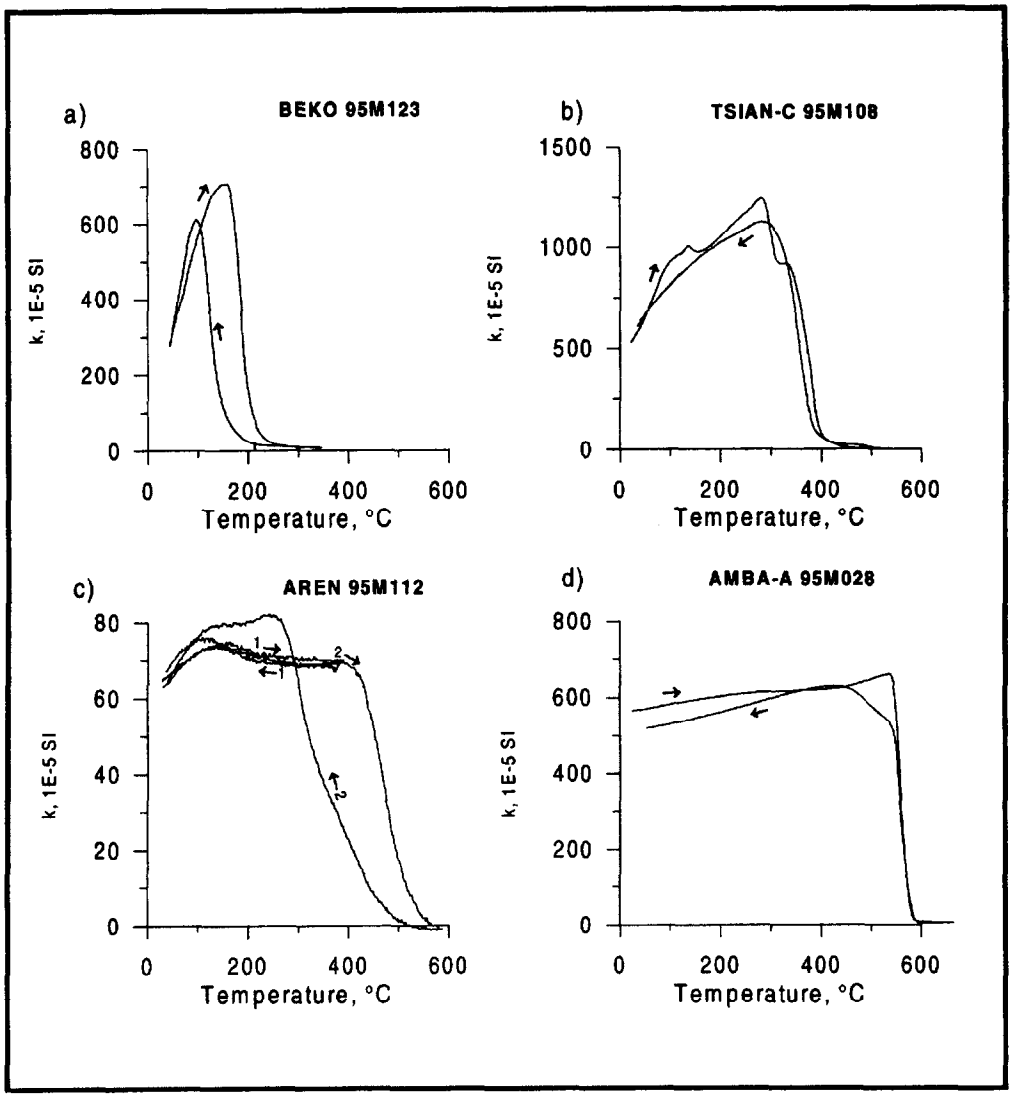


Figure 2. *Examples of susceptibility versus temperature dependencies.*

of significant secondary lightning-induced isothermal remanent magnetisation (LIRM) in many samples. The demagnetisation results of individual samples confirm the presence of secondary LIRM. A typical example of a sample struck by lightning is given in Fig. 4, which shows AF, thermal and LT combined with AF demagnetisation performed on three specimens from the same drill core. Thermal cleaning (Fig. 4a) fails to isolate the primary thermo-remanent magnetisation (TRM), as the broad spectrum of unblocking temperatures of the LIRM component completely masks the primary component, and the expected direction (i.e. the site-mean direction) is not perfectly reached even at high temperatures. AF demagnetisation on a sister specimen (Fig. 4b) is far more effective, as only a field of 20 μT is sufficient to remove the secondary LIRM component completely. Although LT demagnetisation is suggested to be especially effective in removing LIRM residing in multidomain grains (Schmidt, 1993), its effectiveness strongly depends on the mineralogy (e.g. Özdemir *et al.*, 1993; Moskowitz *et al.*, 1998). In general, it was found that only a small proportion of the initial remanence is eliminated after LT demagnetisation (Fig. 4c). For the majority of samples, the authors chose to use only AF demagnetisation to isolate the primary component.

One complete site (Betan-B), and individual samples from some of the other sites (Table 1), had to be excluded from further analysis because the LIRM component completely overprinted the ChRM component. In all other cases, remagnetisation was only partial and could be removed by AF demagnetisation, after which the directions of ChRM could be estimated. Only the samples from two sites were completely unaffected by lightning, namely Aren and Betan-A.

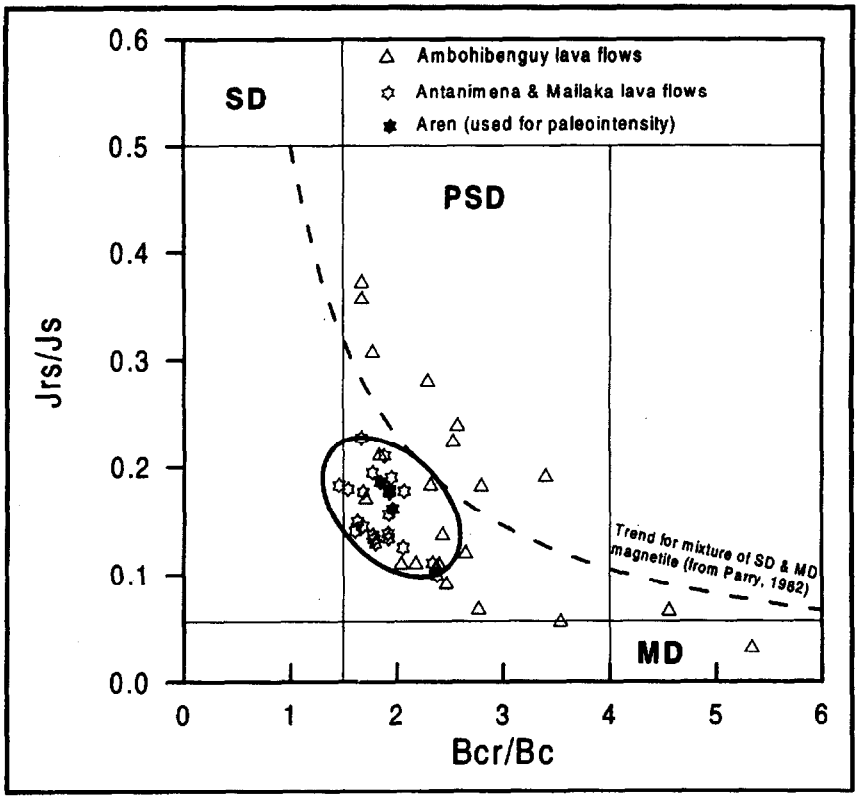


Figure 3. *Day plot (Day* et al., *1977) of the hysteresis parameters ratios.*

Directions of ChRM of the sites sampled south of the Ambohibengy intrusion are generally very scattered and a well-defined mean direction could be found only for two sites, Amba-A and Amba-B (Table 1). As discussed earlier, rock magnetic investigations suggest that these rocks are hydrothermally altered with the main magnetic carrier being secondary pure magnetite with varying grain size. The inconsistent directional palæomagnetic data might, thus, be caused by a long acquisition time for the chemical remanence created during the alteration, or by the new mineralogy being less resistant to lightning. Besides hydrothermal alteration, a rotation of the lava flows during the intrusion emplacement cannot be excluded, and the authors have, therefore, chosen not to consider the palæomagnetic data from the Ambohibengy lava flows.

After excluding all Ambohibengy sites and the lightning struck Betan-B site, only seven out of the original 15 sites could be used in the following analysis. The site-mean directions of these seven sites are given in Table 1. It is noted that all sites have very good within-site statistics with $\alpha_{95} < 3°$ and that they are all of normal polarity, giving a palæomagnetic pole of 74.0°S, 43.7°E (A_{95} = 5.3°, K = 130, N = 7).

PALÆOMAGNETIC POLE AND PALÆOSECULAR VARIATION

In Table 2, all previously published Late Cretaceous palæomagnetic poles for Madagascar are listed, together with their statistical parameters. Based on the reported stratigraphic and K/Ar ages, which range from 68 to 97 Ma (Table 2), the Madagascan palæomagnetic data have traditionally been grouped into different geographic localities having different ages assigned (Andriamirado, 1971; McElhinny and Cowley, 1978; Besse and Courtillot, 1991; Lock and McElhinny, 1991). As noted earlier, the new $^{40}Ar/^{39}Ar$ datings of the Madagascan basalts (Storey *et al.*, 1995) indicate that the Cretaceous volcanism was more or less contemporaneous all over the island, taking place around 88 Ma and having a duration no longer than 6 Ma. It seems straightforward, therefore, to improve the Late Cretaceous palæomagnetic directional data set for Madagascar by combining all the reliable results from all geographic localities of the island.

The palæomagnetic data listed in Table 2 were, however, obtained with rather different demagnetisation and analysis techniques and a careful reselection of the data is necessary to ensure that all data, selected for the combined data set, have a quality comparable to the directional data obtained from this study. The early studies of Roche and

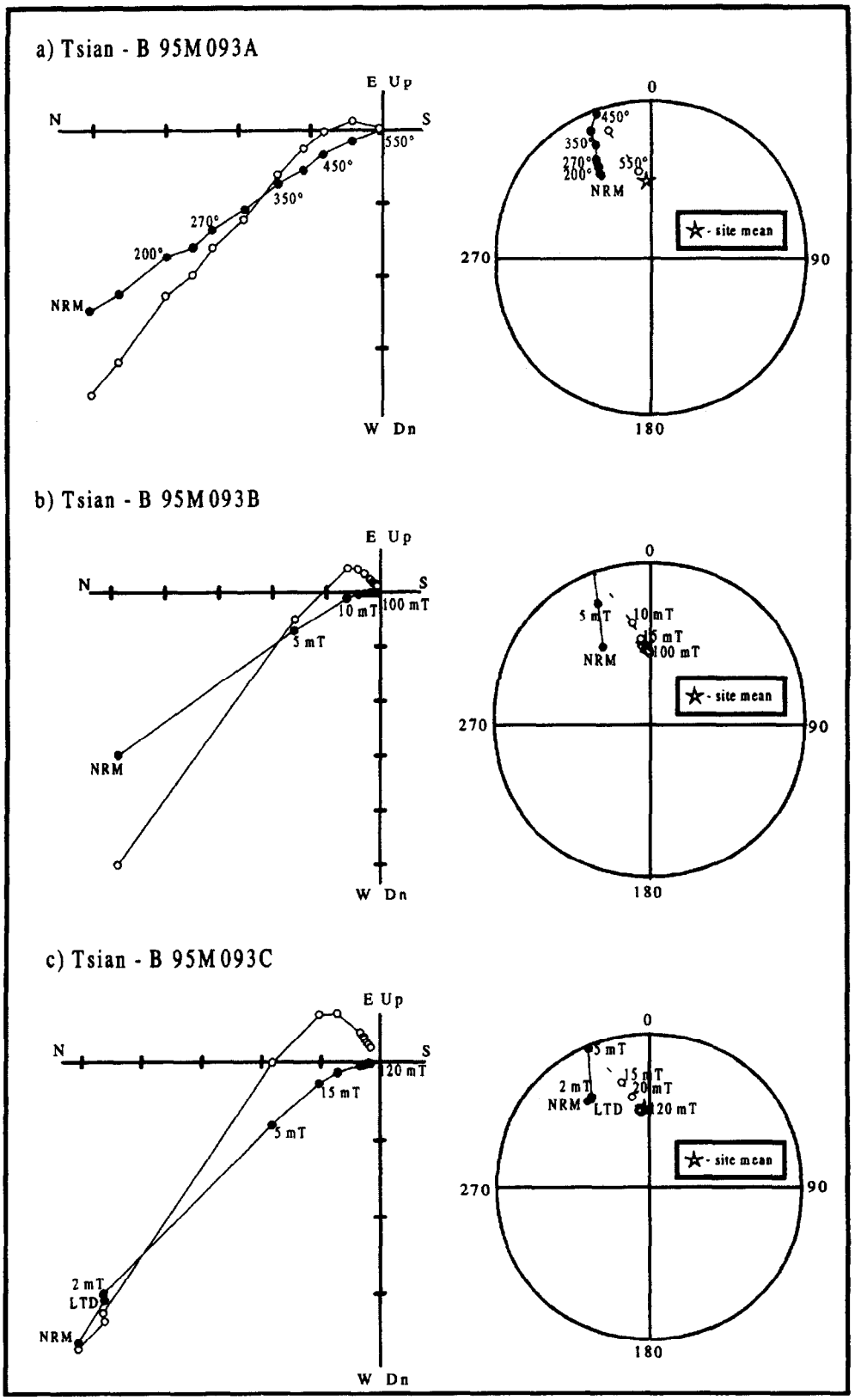


Figure 4. *Examples of orthogonal and equal-area projections of thermal (a), AF (b) and low temperature (c) demagnetisations of three specimens from the same drillcore. On orthogonal projection, ○: vertical planes, ●: horizontal planes; on equal-area projection, ○: negative inclination, ●: positive inclination. The site-mean directions (obtained independently from other samples of the same sites) are shown as stars on the equal-area projections.*

Table 2. Comparison of palæomagnetic poles from previous studies of Madagascan rocks

Rock type and locality	Age	N	VGP Lat °S	VGP Long °E	A_{95}°	K
			Roche and Cattala (1959)			
Lavas and dykes	88–90*	10	66.5	16.5		
			Nairn (1964)			
Lavas	83–97*	2	49.0	352.0	7.6	
			Andriamirado and Roche (1969)			
Volcanics, Mangoky-Onilahy	74–90*	9	79.0	70.0	10.0	30.0
Volcanics, Massif d'Androy	74–90*	7	65.0	72.0	6.0	116.0
			Andriamirado (1971)			
Volcanics, Massif d'Androy	68–76	6	64.0	63.0	7.6	77.7
Volcanics, Southeast Coast	72–77	14	65.8	35.6	4.4	81.5
Dolerites, Tamatave	72–80	10	60.2	32.1	2.8	286.0
Volcanics combined	68–80	30	63.5	39.6		
Volcanics, Mangoky-Onilahy	88–90	11	73.7	73.1	8.9	27.1
Volcanics, Mailaka	88–91	10	70.3	63.1	6.9	49.1
Volcanics Antanimena	88–91	12	66.1	49.7	4.9	78.9
Volcanics combined	88–91	33	69.1	60.1		
			Storetvedt *et al.* (1992)			
Dolerites, Tamatave	~95	16	65.5	38.0		
			Torsvik *et al.* (1998)			
Lavas, Mangoky-Onilahy	~88**	3	76.8	248.2		
			Present study			
Volcanics, Antanimena and M Mailaka	~88**	7	74.0	43.7	5.3	130.1

*: K/Ar or stratigraphic age; **: $^{40}Ar/^{39}Ar$ age; N: number of sites; Lat: latitude of the VGP; Long: longitude of the VGP; K: precision parameter of the Fisher statistics; A_{95}: the confidence cone of the Fisher statistics.

Cattala (1959) and Nairn (1964), for example, seem to be based on rather crude demagnetisation and measuring techniques. As these two studies do not include sufficient information (e.g. within-site dispersion) to allow the authors to assess the reliability of the data, it was decided not to include these data into the combined data set. Though the palæomagnetic work of Andriamirado and Roche (1969), Storetvedt *et al.* (1992) and Torsvik *et al.* (1998) are more recent, site-mean directions are not mentioned, and therefore, these results are not considered either. The directional results of Andriamirado (1971, 1976) are fully published but often sites having only two samples or very low precision parameter (κ) are included in his analysis. For this study, the authors have chosen to use only those sites of Andriamirado which have at least three samples, a precision parameter (κ) higher than 80 and a confidence limit (α_{95}) $< 10°$. Out of his original 63 sites (Table 2), only 37 passed this reselection. Considering the very common occurrence of secondary LIRM components, that were observed in this study (Fig. 4), it is important to note that Andriamirado used only AF to isolate ChRMs.

Combining the seven sites of sampling from this study with the 37 sites of Andriamirado, altogether 44 sites from five separate localities (Antanimena, Mailaka, Côte Centre Nord-Est, Côte Sud-Est, Mangoky-Onilahy; see Fig. 1) were obtained. In Table 1, the geographic coordinates of the sites, together with their site-mean directions, are given. The mean virtual geomagnetic poles (VGP) for each locality are also given in Table 1 and shown in Fig. 5a. It can be seen that the five locality mean VGPs are well grouped, suggesting that no significant local tectonic rotation has taken place on the island since the emplacement of the lavas. Only the mean VGP for Côte Centre Nord-Est (locality 3 in Fig. 5a) is lying a little apart of the other localities, but this is explained by the insufficient number of sites (N = 6) also giving an anomalously low angular standard deviation (*S*) of 3.8 and, hence, clearly indicating an insufficient sampling to average out secular variation. Combining all 44 sites from the five different localities, a palæomagnetic pole for Madagascar with coordinates 67.8°S, 48.5°E ($A_{95} = 3.7°$, K = 34, N = 44) is obtained (Table 1; Fig. 5b).

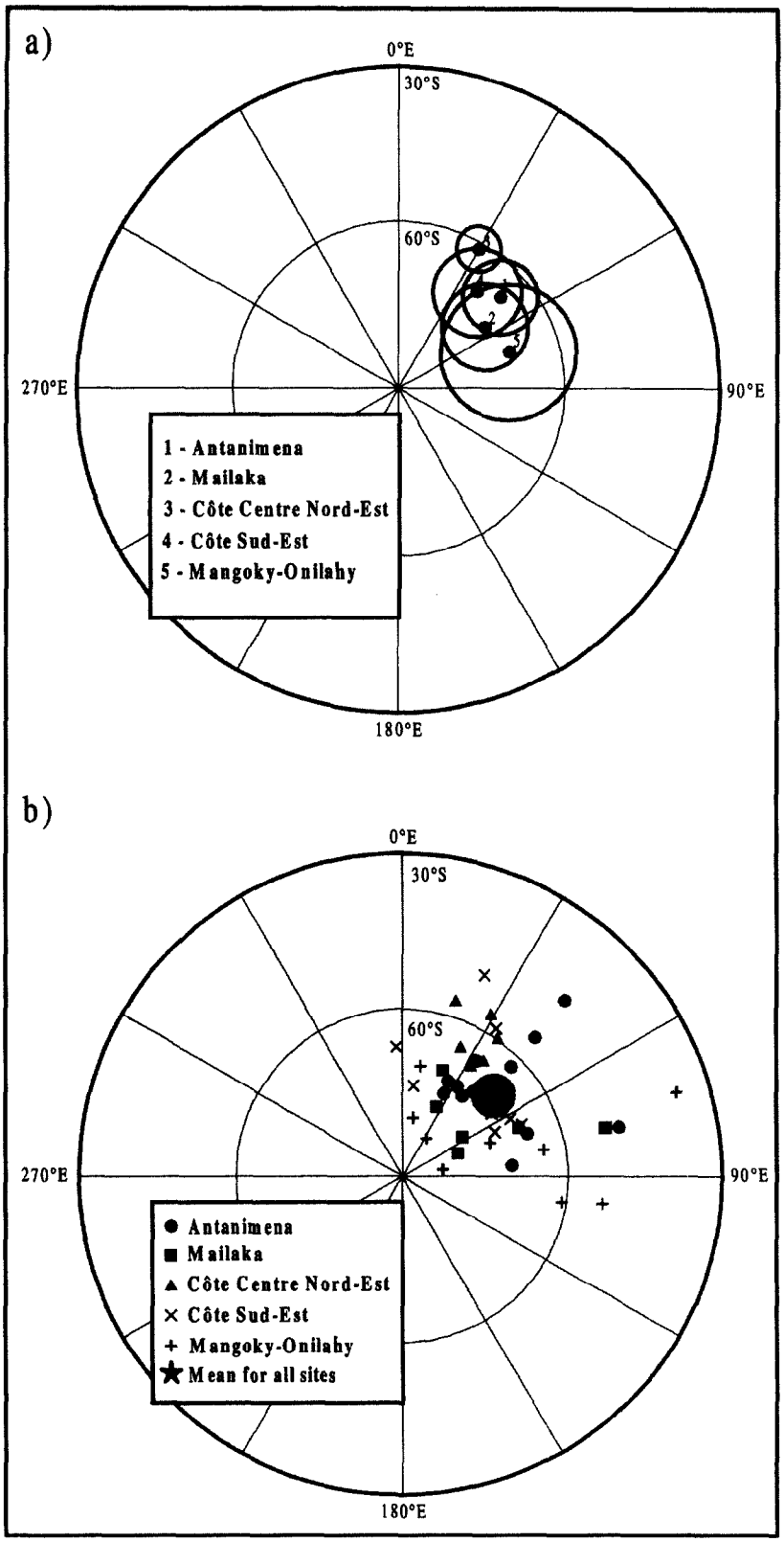


Figure 5. *(a) Equal-area projection of mean VGPs for the five localities with their 95% circle of confidence. (b) Equal-area projection of VGPs for all 44 sites included in this study. The mean VGP for Madagascar (★) is plotted with its 95% confidence limit according to Fisher (circle) and Bingham (ellipse) statistics with the long axis of the Bingham ellipse.*

In Fig. 6, this pole is plotted together with the apparent polar wander path (APWP) for Africa (Besse and Courtillot, 1991), and it can be noticed that it is lying on the APWP within the statistical error limits. In addition to Fisher statistics, Bingham statistics were also applied, in order to evaluate possible elongation of the VGP distribution. Bingham parameters were calculated for every locality and they are given in Table 1. It is worth noting that the mean VGPs calculated by Fisher and Bingham statistics are almost identical, and that the long axis of the Bingham confidence ellipse is directed along the APWP for Africa (Fig. 6).

In order to evaluate palæosecular variation, the authors calculated the S of VGPs distribution. VGPs of all 44 sites were included into the analysis, since using the method of Vandamme (1994) no VGPs were cut off, i.e. no intermediate or erroneous VGPs were included in the studied palæomagnetic data set. Following McFadden *et al.* (1991), S was corrected for within-site dispersion, $S_w = 3.2$, and found to be 13.9 with a 95% confidence limit, $^{+2.4}/_{-1.8}$ (Cox, 1969). This value of S is somewhat smaller than predicted by the PSV model of McFadden *et al.* (1991) for the given age and palæolatitude (Fig. 7). The difference is, however, minor, indicating that secular variation has been sufficiently averaged out, and the palæomagnetic pole represents a time-averaged dipole field.

PALÆOINTENSITY DETERMINATIONS

Following the palæodirectional data and rock magnetic results, only samples from sites showing no presence of lightning induced IRM, little chemical alteration during $\kappa(T)$ experiments and high Curie temperatures (500–550°C) were selected for palæointensity experiments. Unfortunately, this left only two sites and six samples (four from Aren and two from Betan-A) for those experiments. The $\kappa(T)$ curves of these sites are reversible up to ~450°C, while at higher temperatures they are irreversible (Fig. 2c). Palæointensity experiments were performed using the Thellier method in its classic form (Thellier and Thellier, 1959) with all heatings made in vacuum better than 10^{-4} mbar, and laboratory field set to 40 μT. In order to control the occurrence of chemical changes produced by heatings, partial

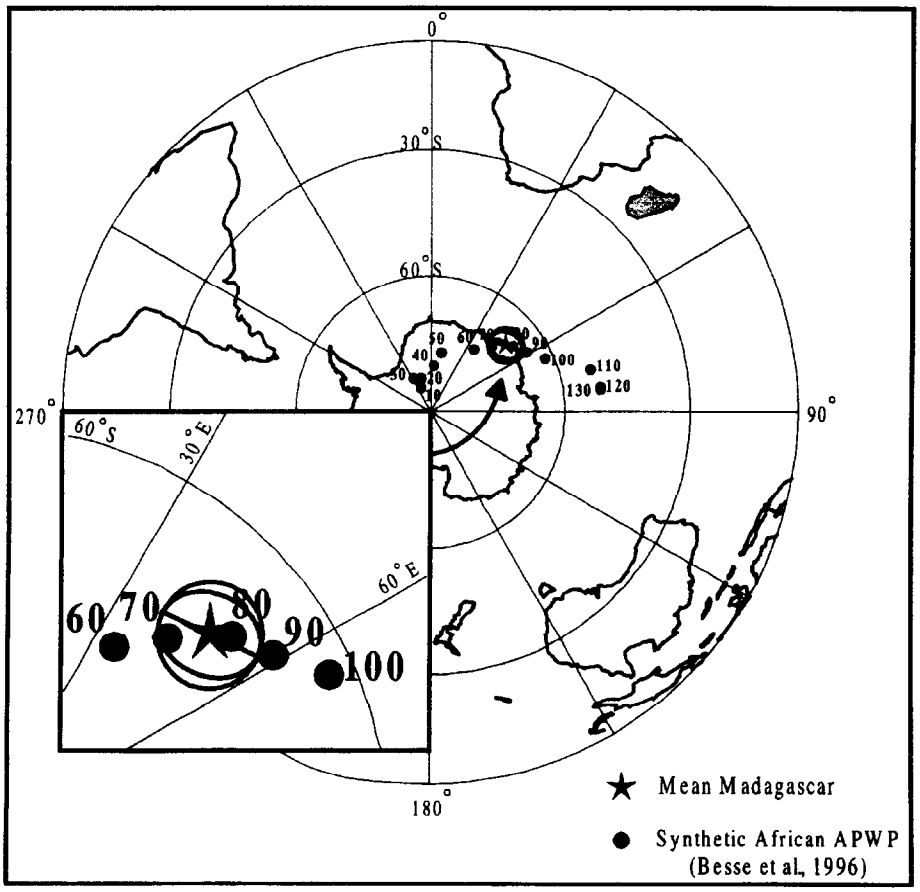


Figure 6. *The palæomagnetic pole for Madagascar, together with Fisher and Bingham 95% confidence limits, plotted on the synthetic APWP for Africa (Besse and Courtillot, 1991). Note that the long axis of the Bingham ellipse lies along the APWP.*

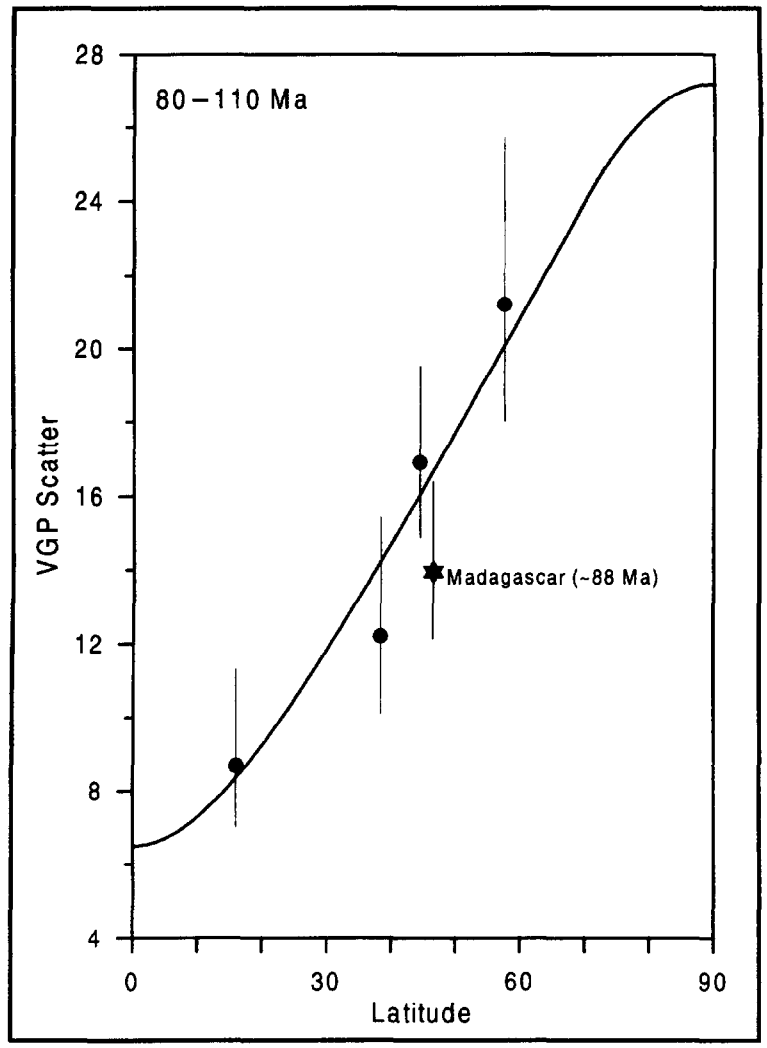


Figure 7. *Palæosecular variation for lavas for 80–110 Ma (after McFadden* et al.*, 1991). The VGP scatter (*$S = 13.9\ ^{+2.4}/_{-1.8}$*) for Madagascar is shown as a diamond.*

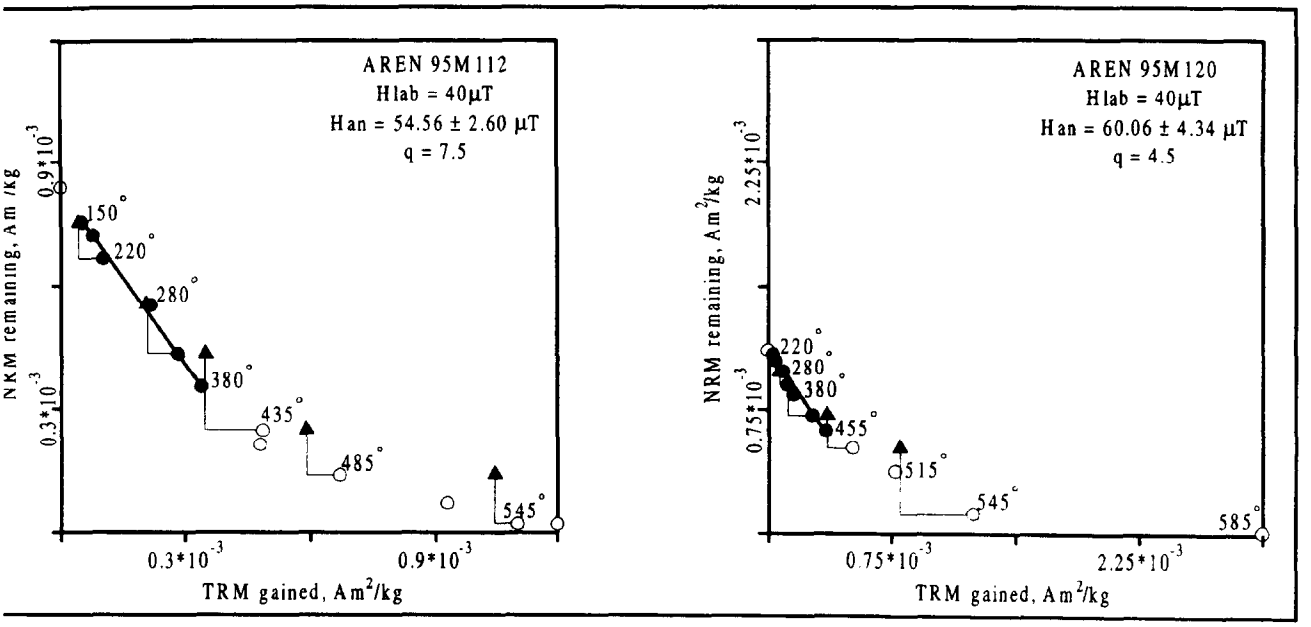
thermo-remanent magnetisation (pTRM) checks were performed after each second heating step.

Samples from Betan-A failed to yield any results, as chemical changes took place in an early stage of the experiment. However, it was possible to estimate palæointensities for all Aren samples, where a linear NRM-TRM relation was observed in the low and medium temperature range (Fig. 8). Corresponding results are summarised in Table 3. The quality factors q associated with the individual determinations are fairly low (< 10) but the small dispersion of the values ($\sigma < 9\%$, where σ is standard deviation) reinforces the reliability of the estimation. The mean value of the palæofield for the Aren site is $61.5 \pm 5.5\ \mu T$, which corresponds to a virtual dipole moment (VDM), calculated at the magnetic palæolatitude of the site, of $10.7 \pm 0.9 \times 10^{22}\ A\ m^2$.

DISCUSSION AND CONCLUSIONS

Palæomagnetic pole and APWP

Combining the new palæodirectional results from this study with those of Andriamirado (1976), the authors obtained a well-defined Late Cretaceous palæomagnetic pole for Madagascar with coordinates 67.8°S, 48.5°E ($A_{95} = 3.7°$, $K = 34$, $N = 44$). The reliability of this pole is strengthened by the fact that the two independent data sets, the mean Madagascan pole obtained by Andriamirado (1976) and the pole obtained from the seven sites from this study (see Table 2), are in good agreement. Another point, adding to the reliability of the combined Madagascan pole, is that no statistical difference is seen in the palæomagnetic poles of the five different localities (Fig. 5a), with a distance between localities as high as 600 km. This fact was taken as an indication that no significant local tectonic deformation has taken place on the



ure 8. Examples of NRM-TRM plots. ●: represent accepted points; ○: represent rejected points; ▲: represent pTRM ecks.

Table 3. Madagascan basalt palæointensity determinations

Site	Sample	$H_a \pm \sigma$ μT	ΔT °C	N	*q*	VDM 10^{22} A m²
Aren	95M112	54.6 ± 2.6	150–380	6	7.5	9.5
	95M120	60.1 ± 4.3	220–455	7	4.5	10.4
	95M121	65.2 ± 8.4	150–435	7	1.8	11.3
	95M122	66.8 ± 5.9	280–455	5	3.3	11.6
Mean		61.5 ± 5.5				10.7 ± 0.9

$H_a \pm \sigma$: either the sample palæointensity with the associated error on the slope or the mean palæointensity with its standard deviation; ΔT: temperature interval used for palæointensity determination; N: number of accepted points: *q*: quality factor (Coe *et al.*, 1978); VDM: Virtual Dipole Moment.

island since the emplacement of the basalts. This is an important observation, as it was difficult at the individual sampling sites to see whether or not the lava flows were tectonically undisturbed. Finally, as already noted, the palæosecular variation has been sufficiently averaged out (Fig. 7), so that the pole corresponds to a time-averaged field.

The oceanic magnetic lineations of the western Somali Basin (Ségoufin and Patriat, 1980; Rabinowitz *et al.*, 1983) clearly indicate that the drift of Madagascar away from Africa had ceased prior to the Late Cretaceous volcanism. This is confirmed by the close agreement of the new palæomagnetic pole for Madagascar with the APWP for Africa (Fig. 6). The fact that the ~88 Ma Madagascan pole falls in the 70–80 Ma interval of the APWP is not statistically significant, because the α_{95} angle of the pole covers the 70–90 Ma part of the APWP. It is noted, however, that three poles from the Early Cretaceous Madagascan basalts with too young K-Ar ages 70, 73 and 74 Ma, respectively, (Andriamirado, 1971) are included in the data set for the synthetic APWP (Besse and Courtillot, 1991; Besse *et al.*, 1996). Hence, the minor discrepancy between the Madgascan pole in this study and the African APWP is probably related to imprecise ages of some of the poles used in generating the synthetic APWP for Africa (Besse and Courtillot, 1991; Besse *et al.*, 1996).

Moreover, the close agreement between the Madgascan pole presented here and the African APWP indicates that the time-averaged Late Cretaceous geomagnetic field was a dipole field. Secondly, if Africa experienced an intraplate deformation in Jurassic-Cretaceous times (e.g. Unternehr *et al.*, 1988), as also was indicated from palæomagnetic data (e.g. Kosterov and Perrin, 1996), this deformation

must have ceased prior to the volcanism in Madagascar. Finally, it is noted that according to the palæomagnetic pole presented here, Massif de l'Androy in southern Madagascar (Fig. 1) had a palæolatitude of 46.5° at 88 Ma. This palæolatitude is in a good accordance with the present latitude of the Marion hot spot (46.7°) and, hence, supports the idea that Massif de l'Androy, which contains the thickest sequence of Cretaceous volcanic rocks exposed on the island, could have been a focal point of the Marion hot spot (Storey *et al.*, 1995, 1997; Torsvik *et al.*, 1998).

To further characterise the palæodirectional data set of 44 independent VGPs, the authors used Bingham statistics and found an small elongation of the VGP distribution, with the azimuth of the long axis (66.4°; Table 1) lying along the APWP for Africa (Fig. 6). This azimuth of the long axis cannot be explained by the pole flattening effect (Cox, 1970), as it differs from the ~90° azimuth that would be produced by transferring circularly distributed (Fisherian) directions into pole space by the geocentric dipole formula. Hence, the elongation of VGP distribution was interpreted as an indication that the duration of the Late Cretaceous volcanic activity on Madagascar was sufficiently long to provide a record of apparent polar wander. This supports the suggestion of Cederquist *et al.* (1997) that the long axes of the confidence ellipse of the Bingham statistics can help to better define an APWP or give additional detail to poorly established APWPs.

Radiometric $^{40}Ar/^{39}Ar$ geochronological data suggest that the entire duration of volcanism on Madagascar was no more than 6 Ma and that, most probably, the volcanic activity ceased first in the north (Storey *et al.*, 1995). Further constraints on the timing are hard to obtain from the palæomagnetic data since the volcanic activity took place within CNS, and, consequently, there are no magnetostratigraphic data. The accuracy of the locality mean VGPs (Fig. 5a) does not allow any age difference to be inferred with respect to a possible few degrees difference in apparent polar wander. Independently of the geochronological method, an estimate of the duration of the volcanic activity can be obtained using the method proposed by Lewchuck and Symons (1995), in which the difference in lengths of α_{1-2} and α_{1-3} axis of Bingham distribution is ascribed solely to apparent polar wander. Using this method, the authors found a duration of volcanic activity on Madagascar of ~5 Ma, which is in good agreement with the $^{40}Ar/^{39}Ar$ geochronological data.

Palæosecular variation

The 80–110 Ma time interval is crucial in the palæosecular variation analysis of McFadden *et al.* (1991), as it is characterised by anomalously low palæosecular variation at the equator compared to all other time intervals. The estimate of palæosecular variation $S = 13.9\ ^{+2.4}/_{-1.8}$, obtained in this study (Fig. 7), is slightly smaller than the value predicted by McFadden *et al.* (1991). The difference is minor; but, if the duration of the Madagascan volcanism was long enough to record APW as indicated by the Bingham statistic, the estimate should overestimate the true palæosecular variation. The data, therefore, support McFadden and Merrill's (1995) idea of significantly reduced palæosecular variation during the CNS.

Palæointensity

Out of the 77 published palæointensity estimates from the CNS, only 12 fulfill the mild selection criteria of Perrin and Shcherbakov (1997), i.e. having at least three determinations per cooling unit with a σ on the palæointensity estimate <10% and excluding transitional data. The VDM of 10.7×10^{22} A m^2, obtained in this study from the single suited flow, fulfills these reliability criteria. It is somewhat higher than other CNS data (Briden, 1966; Bol'shakov *et al.*, 1978, 1981; Bol'shakov and Solodovnikov, 1981; Tunyi, 1986; Pick and Tauxe, 1993; Sherwood *et al.*, 1993; Juárez *et al.*, 1998), which could question the idea of low intensity throughout the entire CNS (Pick and Tauxe, 1993). High VDM close to the end of the CNS is also in accordance with new palæointensity estimates from the Trodos ophiolite (Juárez *et al.*, 1998). Still, the amount of data remains insufficient to resolve variation of the geomagnetic field intensity during the CNS.

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